Little Lab Coats

🎵 Sound Is Vibration 🎵

Physical Science — Sound & Light | Unit 1, Lesson 1 of 3

📚 **Grade:** 1st Grade ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sound & Light 🧪 **Unit:** Unit 1, Lesson 1 of 3

Every sound you've ever heard was made by something vibrating — even your own voice!

📋 **NGSS Standard:**
1-PS4-1 — Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

🎯 Learning Objective

Students will **IDENTIFY** that sound is caused by vibration and **DEMONSTRATE** this by feeling and observing vibrating objects making sound.

🔑 Key Concept

Every sound you've ever heard was made by something **vibrating**.
Vibrations are tiny, rapid back-and-forth movements — too fast to see, but you can feel them!

🔬 The Science

Sound is not magic — it has a physi[illegible]shes and pulls the air around it. Those pushes and pulls travel through the air as **sound waves** until they reach your ears. No vibration? No sound. It's that simple.

This is why when you strum a guitar string, you can see the blur of the string moving AND hear the tone. When you stop the string from moving (by touching it), the vibration stops — and the sound disappears.

Vibration	Pitch
A tight rubber band vibrates fast → high pitch	Short tube vibrates fast → high pitch

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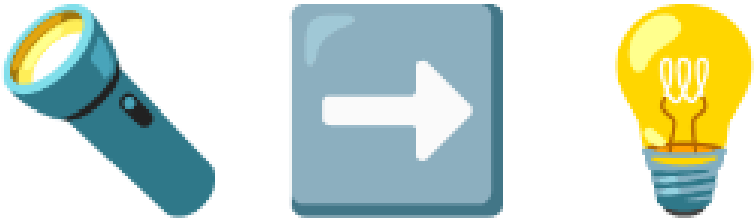
🔦 Light Travels in Straight Lines 🔦

Physical Science — Sound & Light | Unit 1, Lesson 2 of 3

📚 **Grade:** 1st Grade ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sound & Light



Light always travels in a straight line — it can't bend around corners!

📋 **NGSS Standards:**
1-PS4-2 — Make observations to construct an evidence-based account that objects in darkness can be seen only when illuminated.
1-PS4-3 — Plan and conduct an investigation to determine the effect of placing objects made with different materials in the path of a beam of light (transparent, translucent, opaque).

🎯 Learning Objective

Students will **DEMONSTRATE** that light travels in straight lines and **SORT** materials into *transparent*, *translucent*, and *opaque* categories by testing them with a flashlight.

💡 The Big Idea

Light travels in straight lines — always! It cannot bend around corners or curve through tubes. When something gets in the way of light, a shadow forms in the exact shape of the object. And not all materials t[illegible]ight the same way: some let all the light through, some let a little through, and some block it completely.

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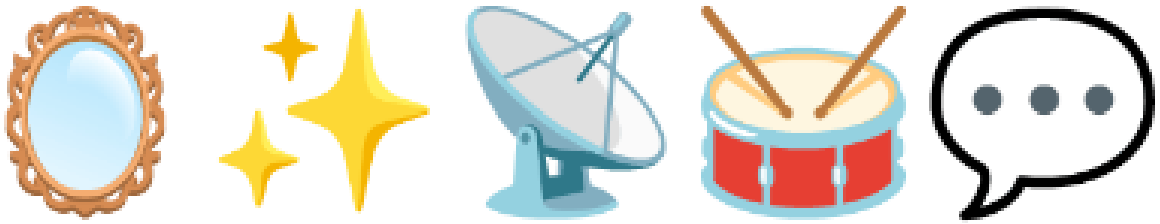
🪞 Mirrors, Signals & Sending Messages 📯

Physical Science — Sound & Light | Unit 1, Lesson 3 of 3

📚 **Grade:** 1st Grade ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sound & Light 📖 **Unit 1, Lesson 3 of 3**



Mirrors flashing in the sun, drums echoing through the forest — humans have always found ways to talk across distances. Today, you design your own!

📋 **NGSS Standard: 1-PS4-4** — Use tools and materials to design and build a device that uses light or sound to solve the problem of communicating over a distance.

🎯 Learning Objective

Students will **DESIGN** and **TEST** a simple device that uses light or sound to communicate a message over a distance.

💡 The Big Idea

Humans have **always** used light and sound to communicate — from drum messages beating across forests to mirrors flashing signals across mountains. Long before phones or radios existed, people were clever engineers, using the tools they had to send messages to faraway places.

Today, your little scientist gets to be that engineer. They'll choose a signaling device, design a simple code, and test whether their message actually makes it across the room!

🔤 **Key Vocabulary:** communicate, signal, vibration, reflect, code, design

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🎵 High Sounds, Low Sounds 🎵

Physical Science — Sound & Light | Unit 1, Lesson 4 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sound & Light

Different vibrations can make different kinds of sounds — some higher and some lower.

📋 **NGSS Standards:**
1-PS4-1 — Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

🎯 Learning Objective

Students will **COMPARE** sounds made by different vibrating objects and **DESCRIBE** some sounds as **higher** or **lower** based on how the object is vibrating.

💡 The Big Idea

Lesson 1 showed that sound is made by vibration. In this lesson, students go one step farther: not all vibrations sound the same. When an object vibrates faster, it often makes a higher sound. When it vibrates more slowly, it often makes a lower sound. Children can notice this with rubber bands, rulers, and their own voices.

> 💡 **Key vocabulary:** vibrate, pitch, high sound, low sound, tight, loose, fast, slow

📦 Materials

🏠 Gather Before You Start

- Empty tissue box or small box with an opening
- 4–6 rubber bands of different sizes or thicknesses
- Pencil or craft stick
- Optional: ruler for a table-edge vibration test

✏️ For the Worksheet

- Printed Student Worksheet (last pages of this lesson)
- Pencil or crayons

🔬 Let's Get Started

Step 1 — Build a Simple Sound Maker (5 min)
Stretch rubber bands across the opening of a tissue box. Pluck one rubber band and watch it move. Ask: *"What do you see the band doing?"*
Remind students that the band is vibrating, and that vibration makes the sound.

Step 2 — Listen for High and Low (6 min)
Pluck one thin band and then one thicker band. Ask your child to decide which sound seems higher and which seems lower. Encourage simple language like: *"This one sounds squeaky and high,"* or *"This one sounds deeper and low."*

💡 **Helpful setup trick:** Slide a pencil under one side of the rubber bands to lift them like little guitar strings. This usually makes the sounds easier to hear.

Step 3 — Tight and Loose Test (8 min)
Gently pull one rubber band tighter and pluck it. Then let it relax and pluck it again. Ask: *"Did the tighter band sound higher or lower?"*
Children should notice that tighter often sounds higher, while looser often sounds lower.

Step 4 — Short and Long Test (8 min)
If you have a ruler, place it on a table with one end hanging off. Pluck the hanging end. Try a short overhang and then a longer overhang. Talk about which sound is higher and which is lower.

🔼 Higher Sound

Often comes from a **faster** vibration, a **tighter** band, or a **shorter** vibrating part.

🔽 Lower Sound

Often comes from a **slower** vibration, a **looser** band, or a **longer** vibrating part.

Step 5 — Try Your Own Voice (5 min)
Say *"hello"* in a high voice and then in a low voice. Ask: *"How can our voices make different sounds too?"*
Connect this back to the idea that our vocal cords vibrate to make sound.

💬 Discussion Questions

🗣️ **Question 1:** "How was this lesson different from Lesson 1?"

💬 *In Lesson 1, we learned that sound is made by vibration. In this lesson, we compare different sounds and notice that vibrations can make sounds that are higher or lower.*

🗣️ **Question 2:** "What change made the sound higher?"

💬 *Strong answers might include making the rubber band tighter, using a thinner band, or making the vibrating part shorter.*

🗣️ **Question 3:** "What made the sound lower?"

💬 *Strong answers might include using a thicker or looser band, or making the vibrating part longer.*

🚀 Extensions

- 🎶 Make a pattern like high-high-low or low-high-low and repeat it.
- 🐦 Listen for high bird sounds outside and compare them to a lower grown-up voice.
- 🥁 Compare the sound of a small container and a larger container when tapped gently.

👩‍🏫 Parent/Teacher Notes

- **This lesson should feel meaningfully different from Lesson 1:** Lesson 1 establishes that sound comes from vibration. Lesson 4 deepens that idea by helping students compare pitch — higher and lower sounds — using the same vibration concept.
- **Keep the language simple:** Grade 1 students do not need formal frequency vocabulary, but hearing "faster vibration" and "slower vibration" is appropriate.
- **Success bar:** students can name a sound as high or low and connect that difference to how the object was vibrating.

🎵 Student Worksheet — High Sounds, Low Sounds

Name: _____ Date: _____

Part 1 — What Did You Hear?

Test	Was it higher or lower?	What changed?
Thin band / thick band		
Tight band / loose band		
Short ruler / long ruler		

Part 2 — Finish the Sentences

A higher sound often comes from a _____ vibration.

A lower sound often comes from a _____ vibration.

Lesson 1 taught me that sound is made by _____.

Part 3 — Draw a High Sound and a Low Sound

Draw two sound makers. Label one higher and one lower.

Draw here ✏️

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Part 1

Answers may vary slightly, but typical responses are:

- **Thin band** = higher, **thick band** = lower
- **Tight band** = higher, **loose band** = lower
- **Short ruler overhang** = higher, **long ruler overhang** = lower

Part 2

- A higher sound often comes from a **faster** vibration.
- A lower sound often comes from a **slower** vibration.
- Lesson 1 taught me that sound is made by **vibrations**.

Part 3

Any sensible drawing of two different sound makers, or one sound maker making a high and low sound, is acceptable.

✅ What Success Looks Like

- Child can describe one sound as higher and another as lower.
- Child can explain that both sounds still came from vibrations.
- Child can point to a change that helped make the sound higher or lower.

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📣 Sound Through Solids, Liquids & Gases 📣

Physical Science — Sound & Light | Unit 1, Lesson 5 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sound & Light

Sound can travel through many kinds of matter — not just air.

📋 **NGSS Standards:**
1-PS4-1 — Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

🎯 Learning Objective

Students will notice that **sound can travel through different materials** and explain that **sound is made by vibrations**.

💡 The Big Idea

Sound needs matter to travel. It can move through air, through water, and through solid objects like tables, walls, spoons, and string. In every case, something starts vibrating, and that vibration moves through the material until it reaches our ears.

> 💡 **Key vocabulary:** sound, vibrate, solid, liquid, gas, air, matter, travel

📦 Materials

🏠 Gather Before You Start

- Plastic cup or bowl of water
- Metal spoon
- Wooden table or desk
- Optional: string and 2 cups for a cup phone

✏️ For the Worksheet

- Printed Student Worksheet
- Pencil or crayons

🔬 Let's Get Started

Step 1 — Sound Through Air (5 min)
Tap a spoon gently on a cup while your child listens from nearby. Ask: *"What is the sound traveling through right now?"* Help them name the air around them. Then ask: *"What was moving or shaking to make the sound?"* Help them notice that the spoon or cup was vibrating.

Step 2 — Sound Through a Table (8 min)
Have your child put one ear gently on a wooden table. Tap the table lightly with your finger or spoon a short distance away. Then listen again with the ear off the table.
Ask: *"Did the sound seem the same or different?"*

💡 **What to notice:** The table is a solid. When you tap it, the table vibrates, and that vibration can travel through the solid table as well as through the air.

Step 3 — Sound Through Water (8 min)
Tap the outside of a bowl or cup of water. Have your child listen beside the bowl while you tap gently. Talk about how the sound is still moving through matter — now including liquid water.

Step 4 — Spoon String Sound Test (10 min)
Tie a piece of string around the handle of a metal spoon. Hold the loose ends of the string and gently swing the spoon so it taps the edge of a table or chair. Listen to the sound with the string hanging free. Then wrap the string ends around one finger on each hand and press those fingers gently against the small flap in front of your ears while the spoon taps again.
Ask: *"When did the sound seem louder or clearer?"* and *"How did the string help the vibration travel to your ears?"*

💨

Gas

Air is a gas. Sound can travel through air.

💧

Liquid

Water is a liquid. Sound can travel through water.

🪵

Solid

Tables, walls, spoons, and string are solids. Sound can travel through solids.

💬 Discussion Questions

🗣️ **Question 1:** "What does sound need in order to travel?"

💬 *Sound needs matter — like air, water, string, or wood — because sound is a vibration moving through something. Ask: "**What did we test that helped prove that?**"*

🗣️ **Question 2:** "Why could you hear the tap through the table?"

💬 *Because the table is a solid, and the **vibration** traveled through the solid material to the ear.*

🗣️ **Question 3:** "What changed when the spoon sound traveled through the string to your ears?"

💬 *The sound usually seemed louder or clearer because the vibration had a direct path through the solid string to the ears.*

🚀 Extensions

- 🚪 Listen at a door or wall when someone taps on the other side.
- 🥄 Try the spoon-string sound test with a wooden spoon and compare it to a metal spoon.
- 🚰 Notice real life examples: hearing splashing in a bathtub, hearing footsteps through a floor, hearing voices through a closed door.

👩‍🏫 Parent/Teacher Notes

- **Keep it concrete:** Grade 1 students do not need vacuum/space explanations here. Just help them notice that sound travels through different materials.
- **Best assessment:** Can the child name at least two materials sound traveled through in this lesson?
- **Safety:** Use gentle taps only. No loud banging near ears.

📣 Student Worksheet — Sound Through Solids, Liquids & Gases

Name: _____ Date: _____

Part 1 — What Did You Test?

Test	What was the sound traveling through?	What did I notice?
Tap heard through table		
Tap heard near water bowl		
Spoon sound through string		

Part 2 — Finish the Sentences

Sound can travel through air, water, and _____.

One solid we tested was _____.

Sound is made by _____.

Part 3 — Draw One Example

Draw one sound traveling through matter.

Draw here ✏️

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Part 1

Accept answers such as: table/wood, water, string, air. Observations may vary. Many children notice the table tap sounds louder or clearer with an ear on the table than with their ear off the table. In the spoon-string test, many children notice the tapping sound seems much louder or clearer when the string helps carry the vibration directly to their ears.

Part 2

- Sound can travel through air, water, and **solids**.
- One solid we tested was **a table**, **string**, or another tested solid.
- Sound is made by **vibrations**.

Part 3

Any sensible drawing showing sound moving through a material is correct.

✅ What Success Looks Like

- Child can name at least two materials sound traveled through.
- Child can explain that sound is vibration moving through matter.

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📣🔦 Sound and Light Message Challenge 🔦📣

Physical Science — Sound & Light | Unit 1, Lesson 6 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 35–45 minutes 🔬 **Subject:** Physical Science — Sound & Light

Sound and light can both carry messages, but each works best in different situations.

📋 **NGSS Standards:**
1-PS4-1 — Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

🎯 Learning Objective

Students will **PLAN**, **TEST**, and **COMPARE** a sound signal and a light signal, then explain which one works best for a communication problem and why.

💡 The Big Idea

This culminating lesson brings the whole unit together. Scientists learned that **sound starts with vibration** and can travel through matter. They also learned that **light travels in straight lines** and can be reflected with shiny surfaces. In this lesson, students use both ideas to solve a real communication problem: when should we use sound, and when should we use light?

> 💡 **Key vocabulary:** communicate, signal, vibration, reflect, light, sound, compare, evidence

📦 Materials

🏠 Gather Before You Start

- 1 flashlight
- 1 small mirror or a square of smooth aluminum foil on cardboard
- 1 metal spoon and 1 cup, pan, or table for making a sound signal
- 2 paper cards labeled **YES** and **NO** (optional)
- A pillow, closed door, or another simple obstacle for one test

✏️ For the Worksheet

- Printed Student Worksheet
- Pencil or crayons

🔬 Let's Get Started

Step 1 — Set the Problem (3 min)
Tell your child: *"Today you need to send a message to someone far away. Sometimes sound is the better tool. Sometimes light is the better tool. We are going to test both and decide."* Review the unit ideas: sound is vibration, and light travels in straight lines.

Step 2 — Make a Sound Signal (8 min)
Choose a simple sound signal, such as one tap for **yes** and two taps for **no**. Have one person tap a spoon on a cup, pan, or table while the partner listens from across the room. Then try again with the listener behind a door or pillow barrier. Ask: *"Could you still hear the message? What was vibrating?"*

💡 **Science connection:** The spoon, cup, table, and air all help carry the vibration. The sound message works because something vibrates and that vibration travels to the listener.

Step 3 — Make a Light Signal (8 min)
Now choose a light signal, such as one flash for **yes** and two flashes for **no**. Shine a flashlight or bounce light from a mirror/foil toward a wall near the partner. Let the partner watch for the flashes and decode the message. Then test again with a barrier blocking the view. Ask: *"Did the light still work when something blocked the path?"*

Signal Detective:
Think about the evidence from both tests.

- Did the signal need a clear path?
- Could it go around a corner or through a door?
- Which signal was easier to notice?

Step 4 — Solve Three Mini Problems (10 min)
Read each situation and let your child choose **sound** or **light**.

- **Problem A:** Your friend is in another room and cannot see you.
- **Problem B:** Your friend is across the yard and can see your flashlight.
- **Problem C:** It is dark, and you want someone to notice where you are.

After each choice, ask: *"What evidence from our tests helps you decide?"*

Step 5 — Share the Best Tool (5 min)
Have your child explain which kind of signal was best for each problem. Encourage them to use the words **vibration**, **light**, **blocked**, and **travel**.

💬 Discussion Questions

🗣️ **Question 1:** "When was sound the better signal?"

💬 *Sound can be better when the other person cannot see you, such as behind a door or around a corner.*

🗣️ **Question 2:** "When was light the better signal?"

💬 *Light can be better when the other person is far away and has a clear line of sight to the signal.*

🗣️ **Question 3:** "Why do sound and light not always work the same way?"

💬 *Sound is vibration traveling through matter. Light travels in straight lines and can be blocked if something is in the way.*

🚀 Extensions

- 🏠 Invent a new code with three messages, such as **come here**, **wait**, and **help**.
- 🌙 Try the light signal outdoors at dusk with adult help.
- 🚪 Test whether a thicker door or pillow changes how well the sound signal works.

👩‍🏫 Parent/Teacher Notes

- **This lesson is the unit capstone:** students are not building another cup phone. They are comparing **two different ways** to communicate and choosing the best tool for each situation.
- **Ideal explanation:** "I chose sound when my partner could not see me, and I chose light when my partner could see the flashes from far away."
- **Success bar:** students should use evidence from the tests, not just guess their favorite signal.

📣🔦 Student Worksheet — Sound and Light Message Challenge

Name: _____ Date: _____

Part 1 — Test Two Signals

Signal	How did we send it?	Did the message work?	What did I notice?
Sound			
Light			

Part 2 — Pick the Best Tool

Circle the better tool for each problem.

Problem	Better Tool	Why?
My friend is behind a door.	Sound / Light	
My friend is far away and can see me.	Sound / Light	
I want someone to notice me in the dark.	Sound / Light	

Part 3 — Finish the Sentences

Sound works because something is _____.

Light works best when it has a clear _____.

The best signal for a hidden partner is _____ because _____.

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Part 1

Accept sensible notes about each test. Strong answers mention that the sound signal could still work when the partner could not see, while the light signal needed a clear path.

Part 2

- **Behind a door:** usually **Sound**
- **Far away and can see me:** usually **Light**
- **Notice me in the dark:** often **Light**, though some children may defend sound with evidence; accept well-supported answers.

Part 3

- Sound works because something is **vibrating**.
- Light works best when it has a clear **path**.
- The best signal for a hidden partner is **sound** because **the partner can hear it even if they cannot see me**.

✅ What Success Looks Like

- Child compares both kinds of signals.
- Child uses evidence from tests to match a signal to a problem.
- Child explains the choice using vibration and straight-line light ideas from the unit.

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